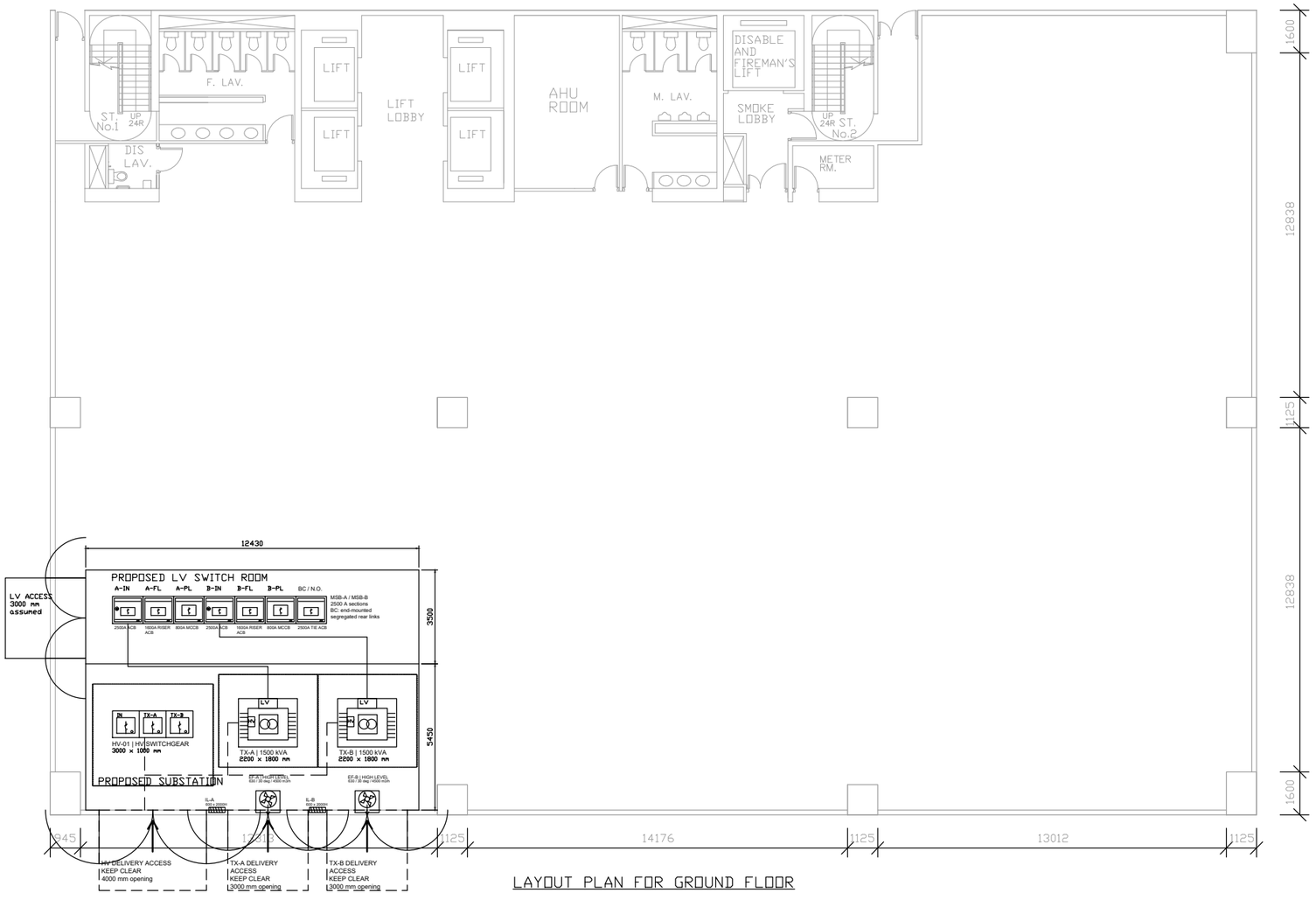
TASK 1 | GROUND-FLOOR PLANTROOM LOCATION

A3 LANDSCAPE | SCALE 1:200 | DIMENSIONS IN mm | GROUND FLOOR



LAYOUT PLAN FOR GROUND FLOOR

PROPOSED LOCATION

Lower-left perimeter of supplied ground-floor plan. Substation 12430 x 5450; adjacent LV room 12430 x 3500 assumed. See separate 1:75 room detail.

LOCATION REASONS

Direct external plant access is proposed, subject to site confirmation.

Adjacent LV room keeps transformer-to-incomer interfaces short.

Perimeter plantrooms separate supply equipment from the upper lift/stair/AHU core and remaining accommodation.

LV room has its own proposed west entrance.

LOAD ASSESSMENT

Offices: 1644.800 kVA.

AC: 1000 kW cooling -> 526.039 kW electrical -> 584.488 kVA at assumed PF 0.90.

Public services: 200 kVA aggregate allowance.

Two-zone conservative envelope: 2672.216 kVA.

Propose 2 x 1500 kVA, with equal zone allocation; no full-load N-1.

SITE CONDITIONS TO VERIFY

External apron, road and <=20m unloading route are not given by the source plan. Maintain >=3m clear delivery route.

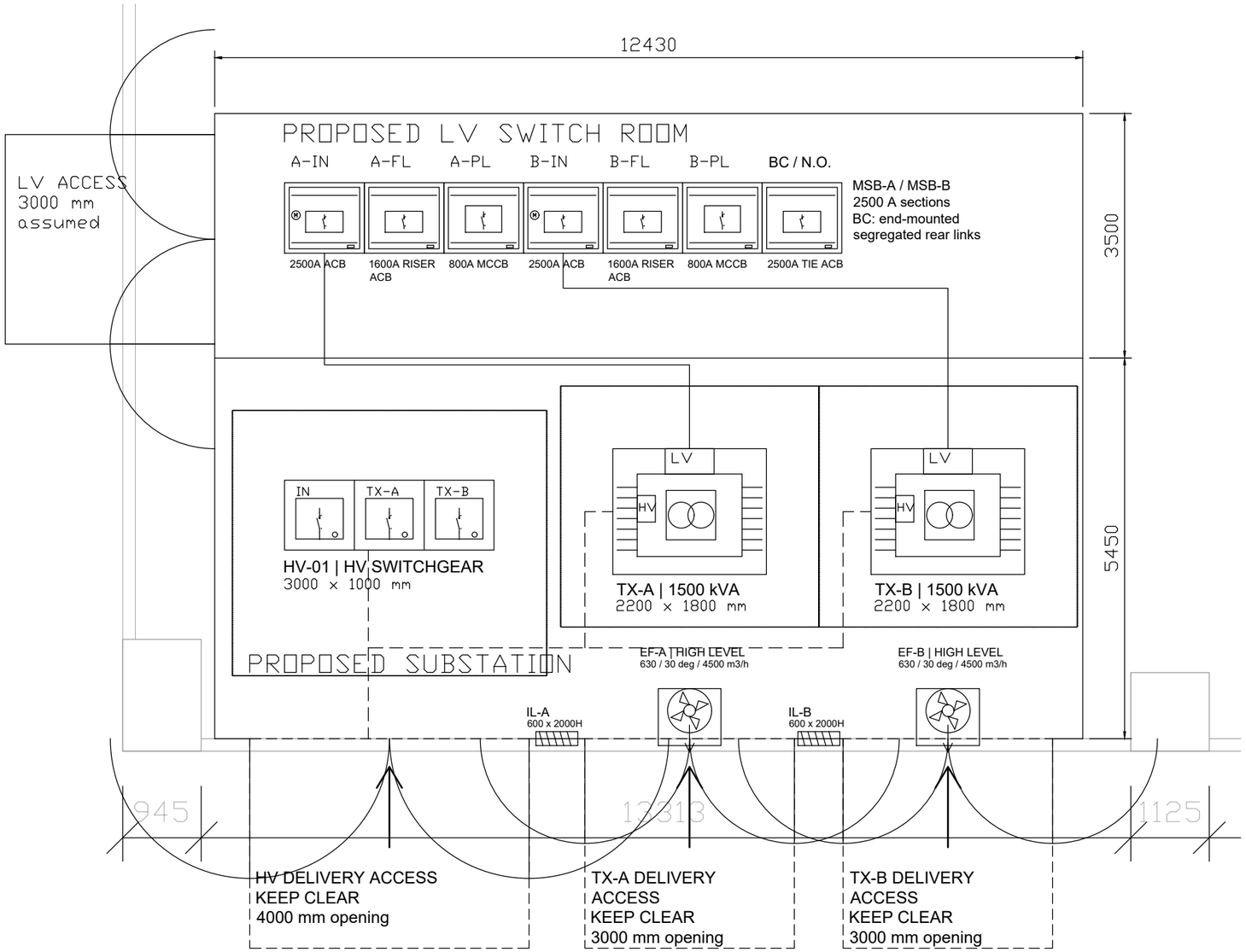
Architectural wall openings, structural/flood/fire provisions, ventilation and supplier equipment approval require coordination.

GROUP 1 | TASK 1

Design basis: CLP COP215 Rev07; CLP COP101 v16.

TASK 1 | PROPOSED SUBSTATION AND MAIN LV ROOM

A3 LANDSCAPE | SCALE 1:75 | DIMENSIONS IN mm | PLANTROOM DETAIL



LINE KEY: dashed path = HV trench; solid path = LV route. Dashed boxes = delivery clearance; arrows = delivery direction. Equipment details diagrammatic.

LV INTERFACES / HV ROUTES

LV trench: 1000 W x 1100D, four MCTs per interface. A: 8.03+2=10.03m; B: 8.31+2=10.31m (<20m). 2m is assumed total termination/vertical allowance.

HV centre-lines: 1000 W at panels; 600 W branches; final 300 W; generally 1400D. Segregate HV/LV. Covered trenches; no beam obstruction.

DETAIL 1:75 AT A3

CAPACITY AND EQUIPMENT

TX-A/B: CLP 1500 kVA; 3000 kVA installed, 2672.216 kVA demand. No full-load N-1.

HV-01: incomer + two transformer protection panels.

MSB: seven 1140 mm bays; 2500 A bus sections. BC: 4P normally-open end coupler, segregated insulated rear links to A/B.

A-IN/B-IN/BC electrically and mechanically interlocked. Transfer after isolation and load shedding.

Installation envelopes are design assumptions, not certified manufacturer dimensions.

CLEARANCES / BUILDING WORKS

HV rear1000, front1800, sides750. TX LV terminals north:900; other sides750.

LV board: front1500, rear990; seven bays x1140 =7980 mm.

Proposed FRR partitions; fire-seal penetrations. FFL +0.15 m vs external +0.00 assumed.

Actual structure and equipment dimensions require coordination.

ACCESS AND DOORS

South external access assumed. Provide 3000-wide clear delivery route; unloading distance <=20 m to verify.

Plant doors: HV 4000 W; TX 3000 W each; 2800 H assumed. Outward opening, shown at 90 degrees.

Adjacent plant doors used sequentially; TX-A has wicket personnel access.

Independent LV entrance west.

VENTILATION

EF-A/B: COP101 v16 6.6 type 630 mm / 30 deg / 700 rpm; each 4500 m3/h at 70 Pa, <=58 dB(A) at 1 m.

Design heat: 2 x 15 kW x 1.25 = 37.5 kW. Combined 9000 m3/h gives 12.44 K air rise.

IL-A/B: each 600 W x 2000H, minimum 50% free area; combined 1.20 m2 >=1.12 m2.

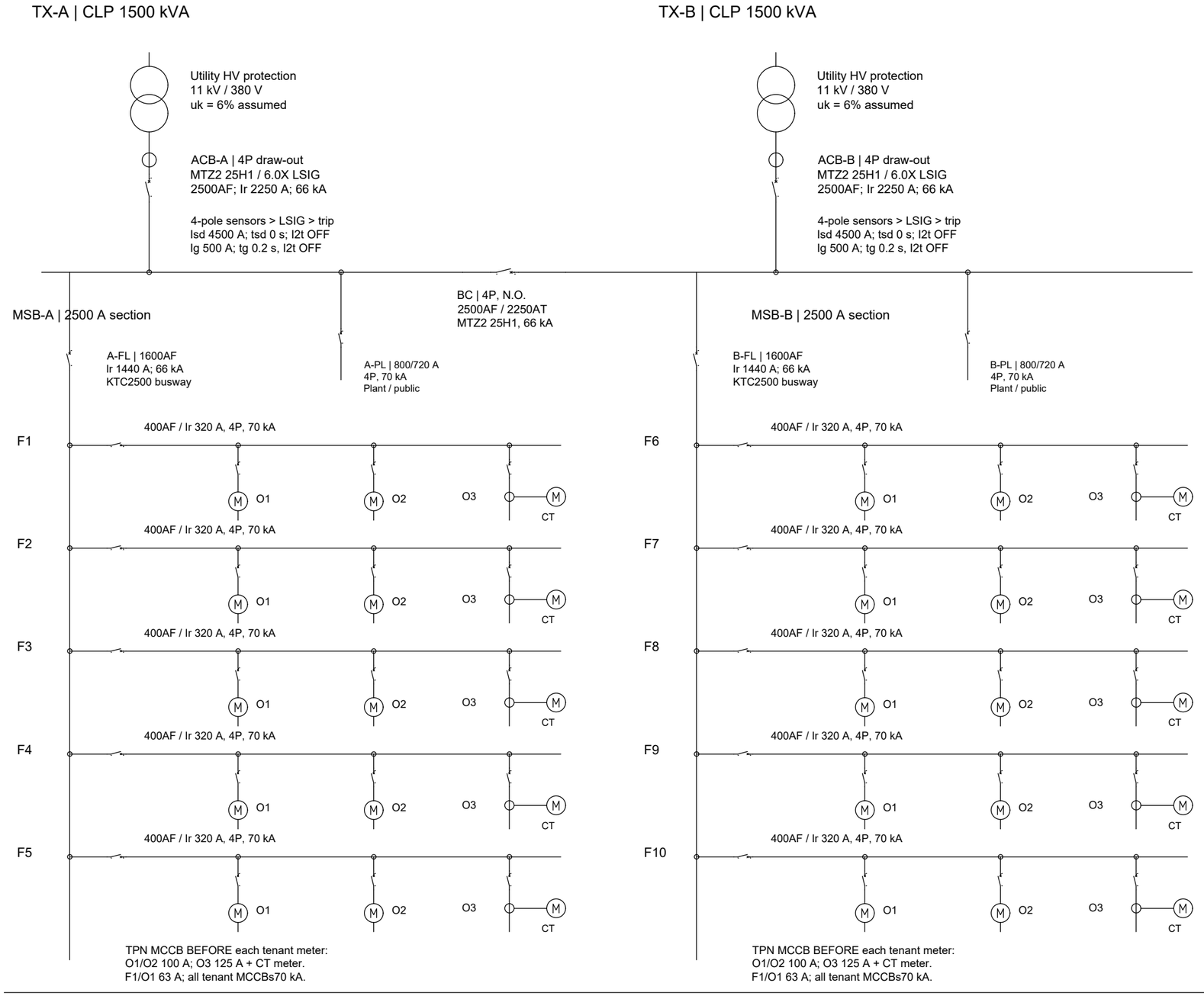
High-level ducts: 900 x 750 mm; discharge >=2.5 m above street, away from people. See ventilation calculation.

CLP COP215 Rev07; COP101 v16, 4.3 and 6.1-6.7.

Equipment selections and calculations accompany this drawing.

TASK 1 | MAIN DISTRIBUTION, PROTECTION AND METERING

REV03 | 380/220 V, 3-phase 4-wire + PE | Single-line diagram, not to scale | Normal supply only



F1 / OFFICE 1 | DETAILED FEEDER

From F1 320 A floor MCCB
MS-O1: ComPacT NSX100H
C10H4TM063, 4P4D TM63D
Ir 63 A; Icu/Ics 70 kA
Suitable for isolation

CLP tenant meter
Immediately after MS-O1

4C35 mm2 PVC Cu
+ separate CPC16 mm2
L = 31.5119 m
28.5119 plan + 3 m allowance

DB-O1: 100 A 4P isolator
15 x 1P+N Type A 30 mA RCBO
7 lighting + 8 power circuits

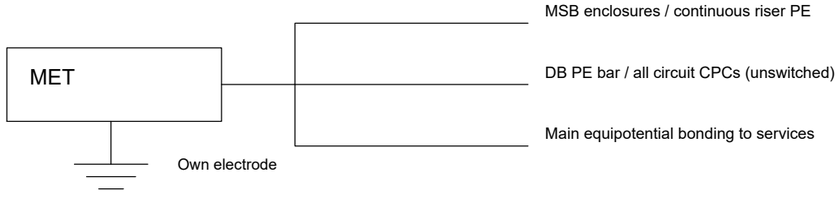
PE continuous; neutral switched with phases.
No customer N-PE link at DB.

ACB-A / ACB-B / BC INTERLOCK

Electrical AND mechanical interlock
Normal: CLOSED / CLOSED / OPEN
Transfer from A: CLOSED / OPEN / CLOSED
Transfer from B: OPEN / CLOSED / CLOSED
ALL THREE CLOSED: PROHIBITED

Open/isolate failed incomer before closing BC.
Before transfer, shed load to <=1480.9 kVA
(2250 A at 380 V); no full-load N-1.
Utility return: open BC before restoring incomer.

CUSTOMER EARTHING



BASIS AND FAULT DUTY

CLP GuideToSupply v10: sections 2, 3, 7.4, 11.
MSB PFC <=56.92 kA incl. transfer motors.
Own MET/electrode; any CLP bond only by agreement, >=70 mm2 Cu. See Task 3 for loop, disconnection and conductor checks.
CT meters: CLP ratios +16 A HRC voltage fuses; Guide sections9.11 and9.16.

2672.216 kVA building demand | 1336.108 kVA / 2030 A per normal section | CT/sensor arrangement subject to CLP acceptance